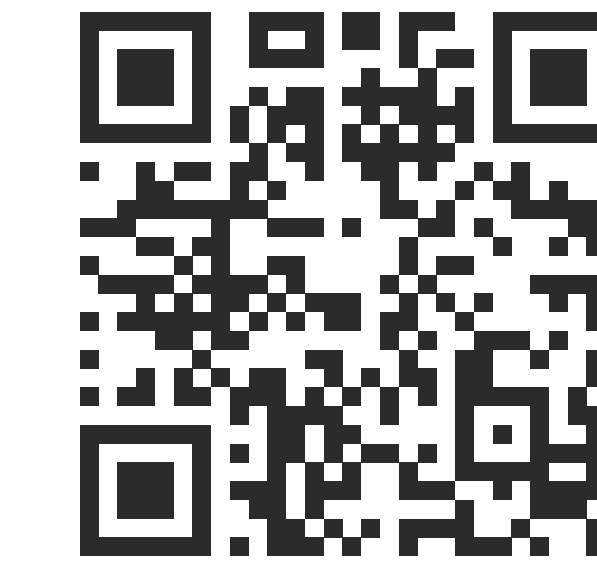


Bootstrapped Watershed

Towards Few-Shot Instance ZooScan Segmentation

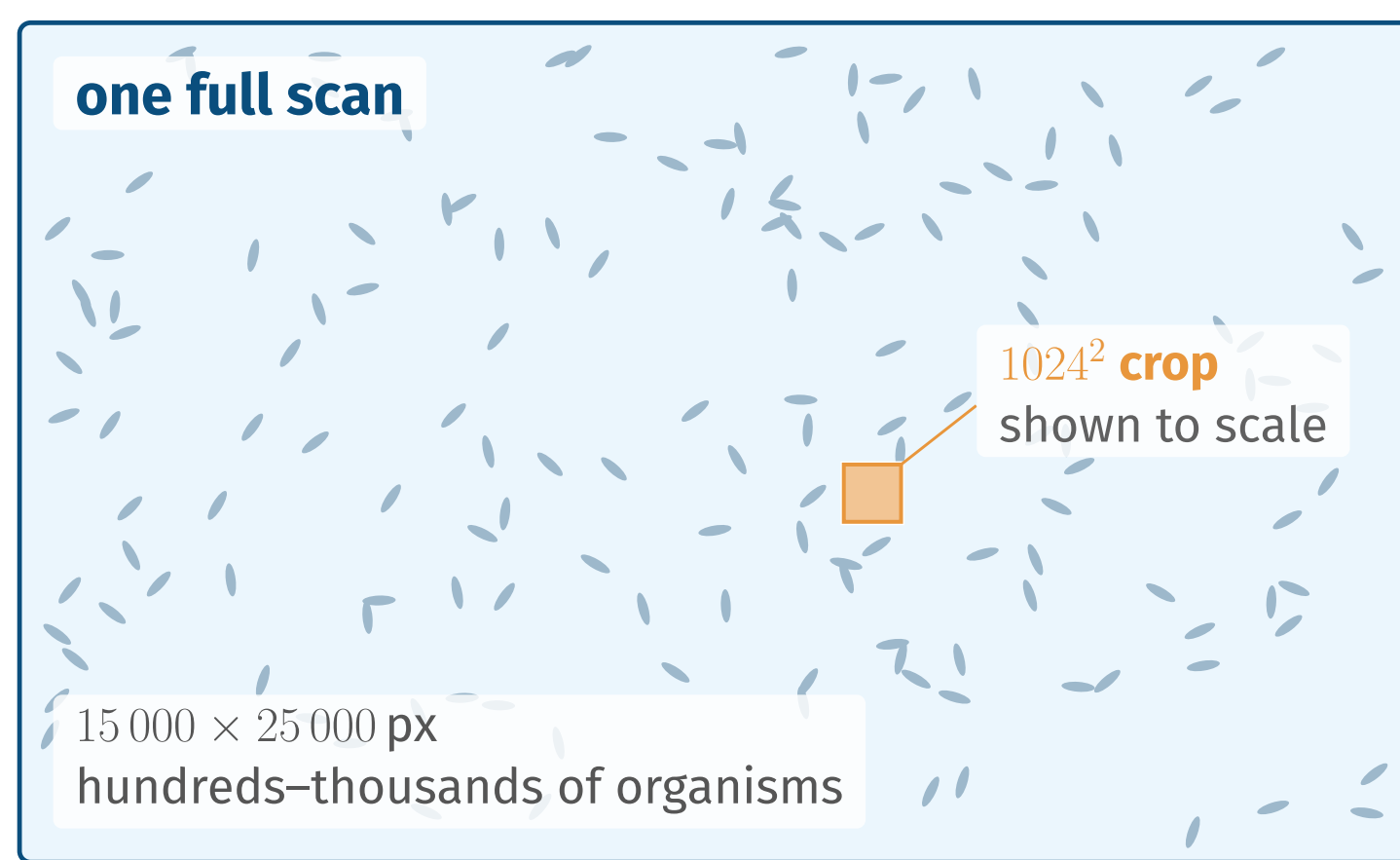
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Dense ZooScan instance segmentation learned from **three annotated crops** — frozen DINOv3 features are bootstrapped into pseudo-labels, then a watershed stage recovers **individual organisms**.

The problem

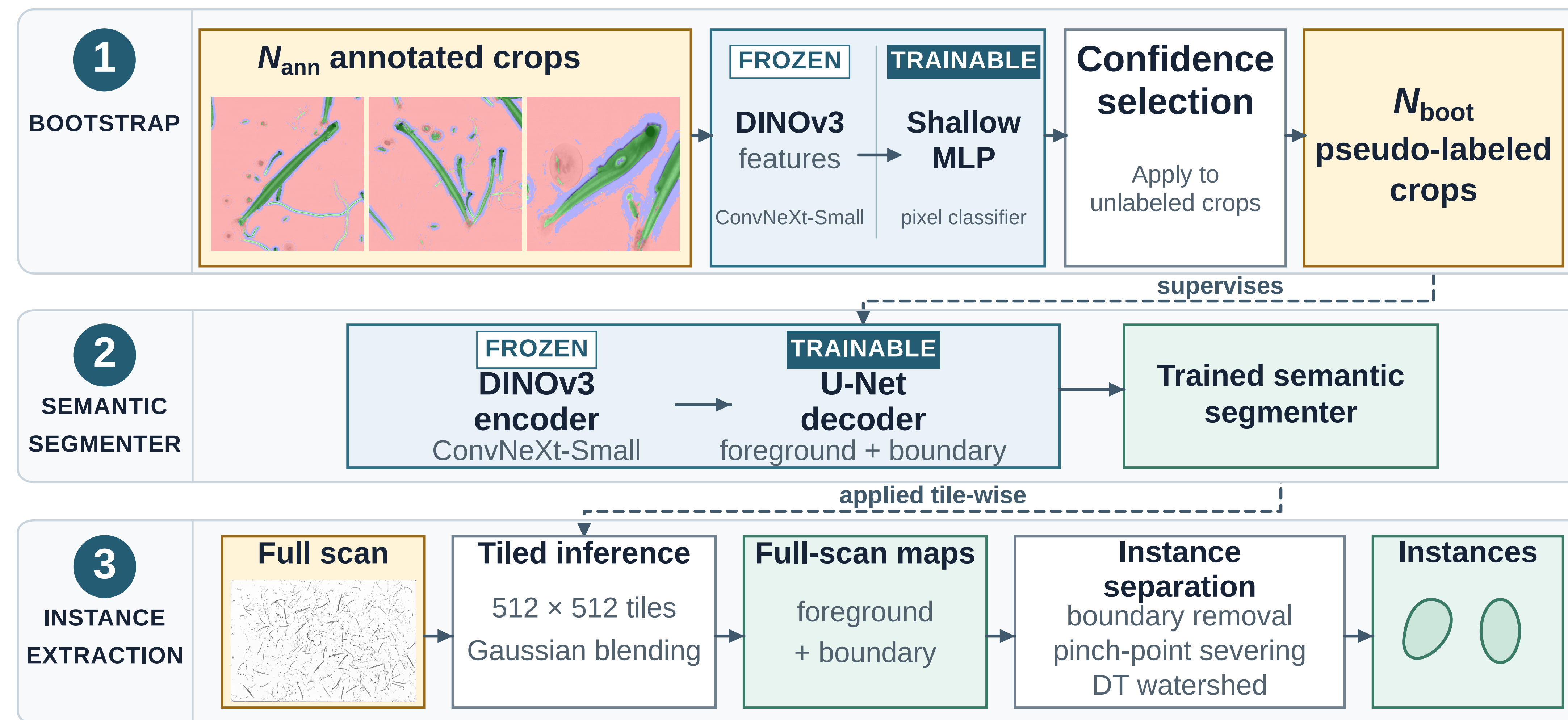
Herring larvae surveys underpin Baltic fisheries stock assessment, but samples are still identified, counted and measured **by hand**. ZooScan digitises an entire sample in a single scan, yet every organism must still be **located and separated from its neighbours** before it can be counted or measured.



images provided by THÜNEN

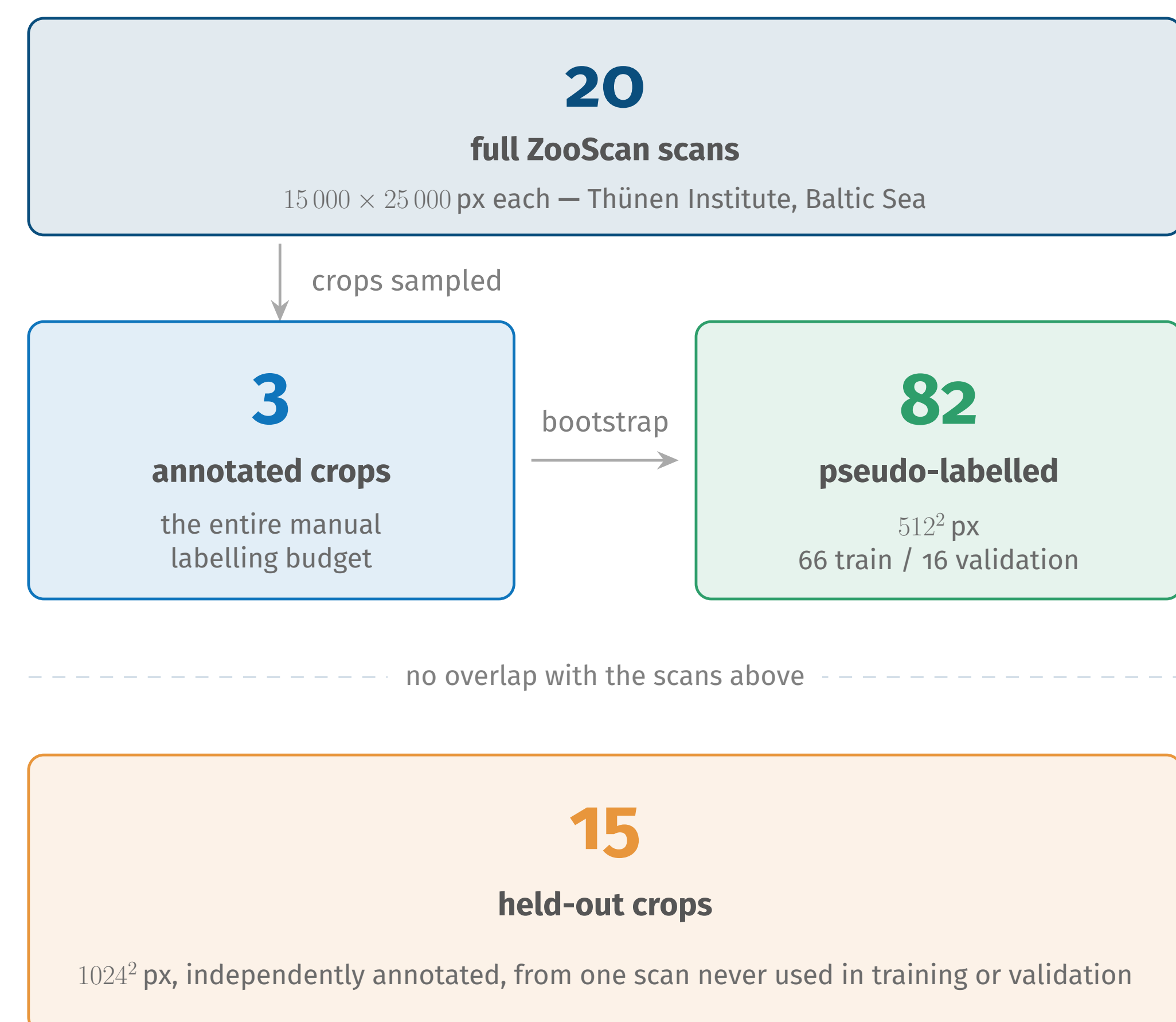
Classical ZooScan / ZooProcess pipelines threshold the scan and take connected components, so touching organisms merge into a single object. Learned instance segmentation would fix that, but needs far more annotation than this domain can afford: few-shot benchmarks presuppose a large annotated **base-class dataset** of small images containing few objects, and no ZooScan equivalent exists. We therefore ask how far **three annotated crops** can be stretched.

Method



Frozen DINOv3 features train a shallow MLP that generates candidate masks for unlabeled crops; the most confident are kept as pseudo-labels. Overlapping tile predictions are then blended into full-scan foreground and boundary maps, from which organism instances are recovered.

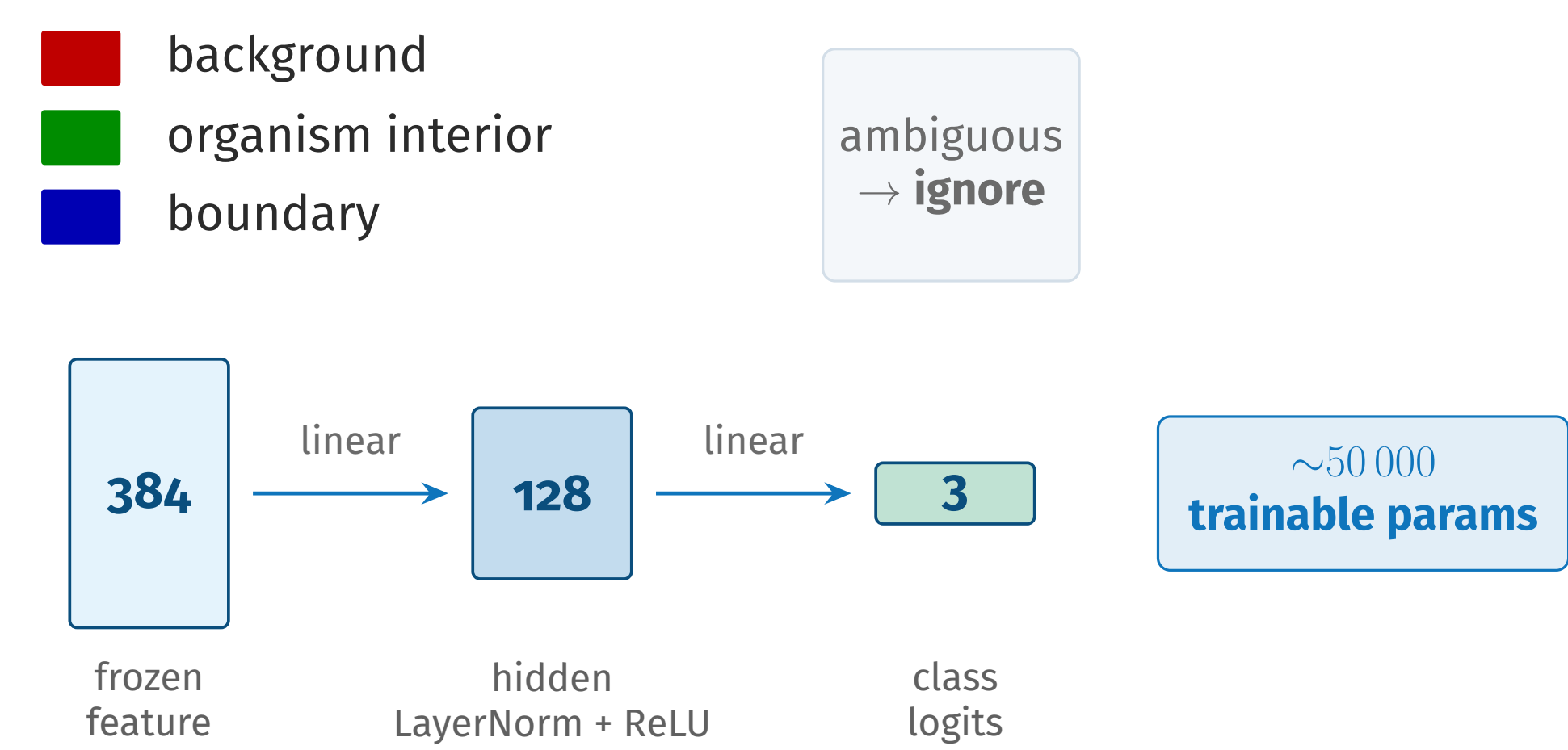
Data



Every visible organism is labelled as a separate instance, including organisms that touch or overlap. All methods are tested on the same 15 crops, from a scan **never seen** in training.

Bootstrap & training

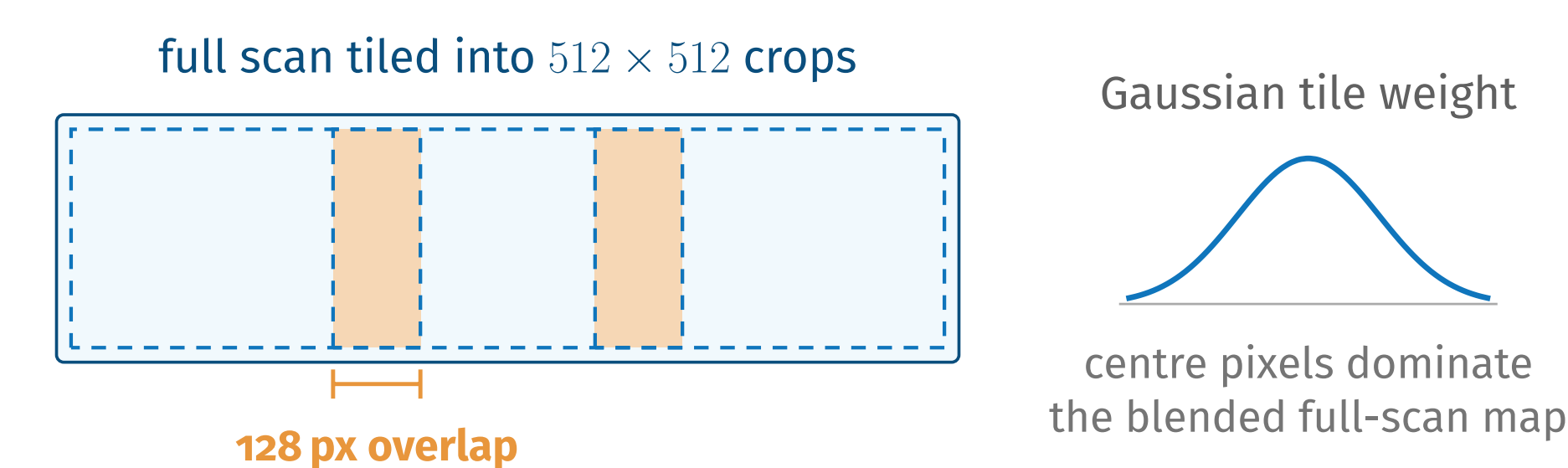
Each annotated crop is labelled per pixel into three classes; ambiguous pixels are excluded from the loss.



- A frozen **DINOv3-ConvNeXt-Small** backbone gives a 384-d descriptor per position; only the shallow MLP above is trained.
- It labels unlabelled crops, and the $N_{\text{boot}} = 82$ most confident 512² crops are kept as pseudo-labels — a 27-fold expansion of the $N_{\text{ann}} = 3$ manual crops.
- These supervise a **U-Net decoder** on the frozen DINOv3 hierarchy: Tversky loss ($\alpha = 0.7, \beta = 0.3$) plus cross-entropy weighted 1.0/8.0/1.5 over background, boundary and interior.

Instance extraction

A scan of $\sim 15\,000 \times 25\,000$ px will not fit in GPU memory, so the semantic maps are predicted tile-wise and blended back together.



The blended foreground and boundary maps become instances in three steps:

- Boundary removal.** Predicted boundary pixels are subtracted from the foreground, separating organisms the network already divided.
- Pinch-point severing.** Pixels with $0 < D_{L2} < 12$ px that carry boundary predictions on ≥ 2 of 4 sides within an 8 px radius are cut; the mask is closed (5×5) and a dilated boundary wall subtracted, so predicted boundaries act as hard separators.
- Distance-transform watershed.** Markers are placed at maxima of the 9×9 Gaussian-smoothed distance transform above 5 px; instances under 150 px² are discarded.

Results

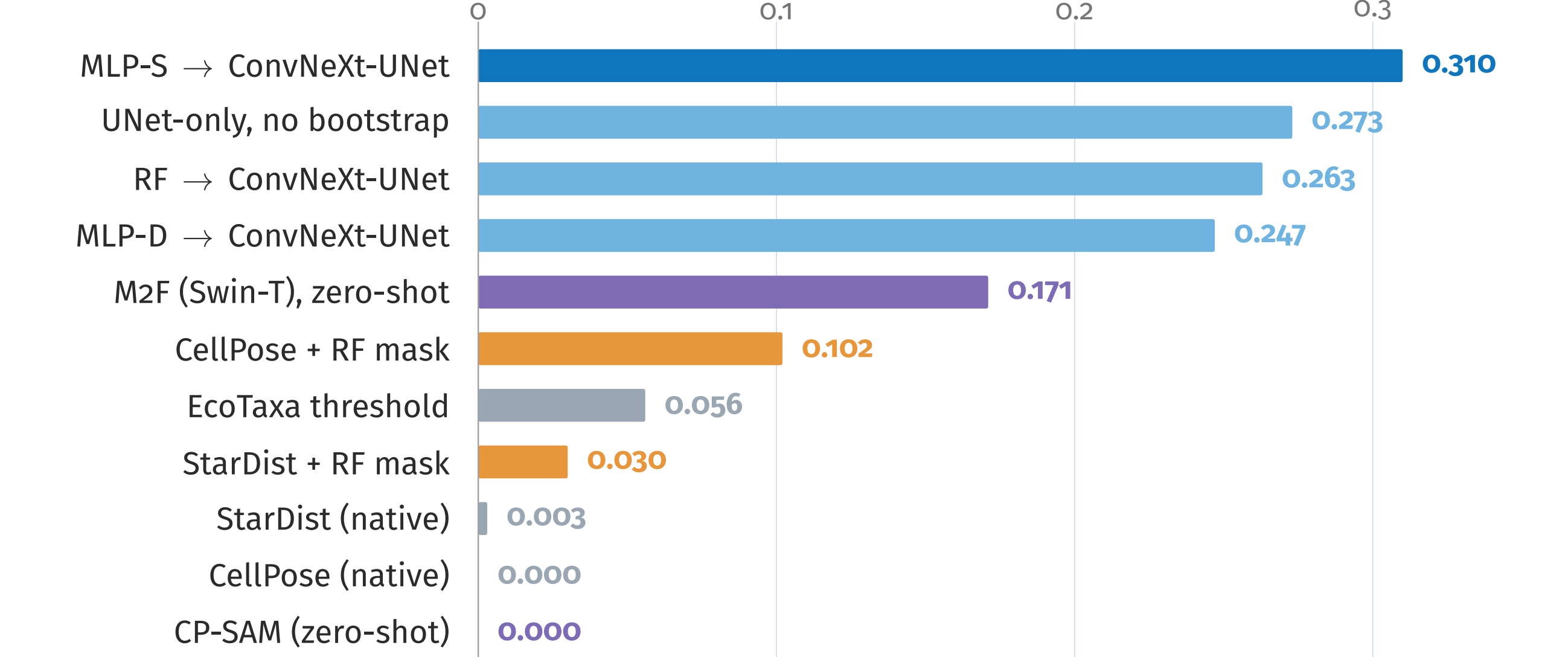
0.310
Panoptic quality
0.273 direct U-Net
0.171 best zero-shot

0.573
Instance recall
0.538 direct U-Net
0.202 best zero-shot

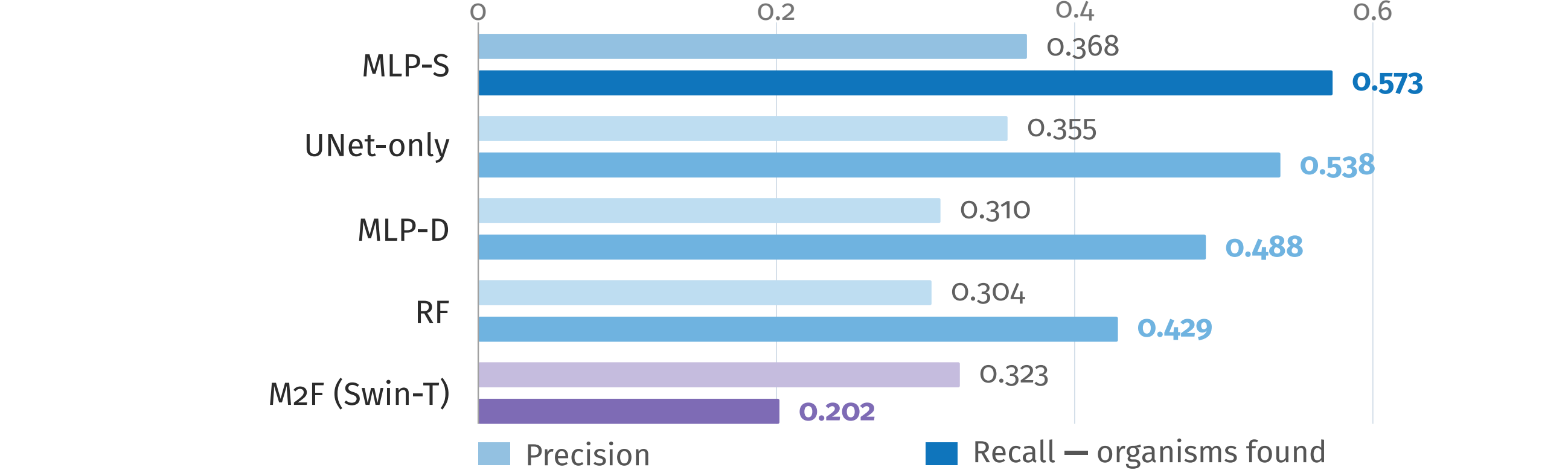
$$PQ = \frac{\sum_{(p,g) \in TP} \text{IoU}(p,g)}{|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|}$$

Macro-averaged over the 15 held-out crops, with predicted and reference instances matched one-to-one at IoU 0.5.

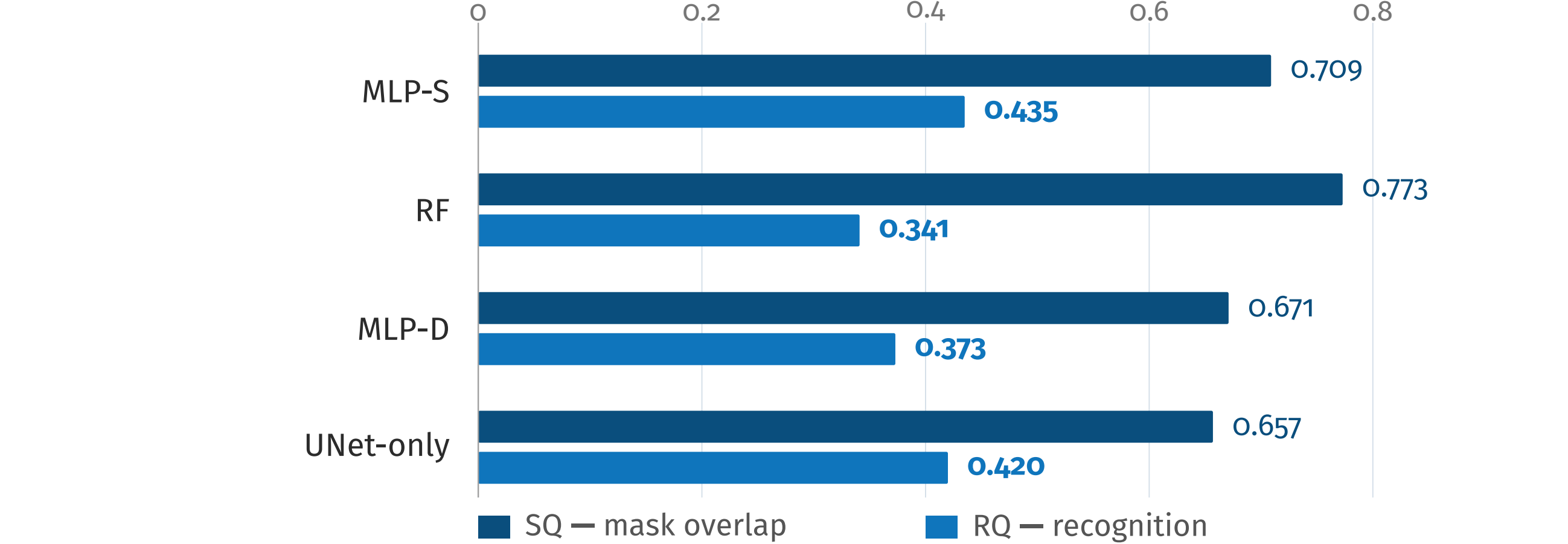
Panoptic quality — higher is better



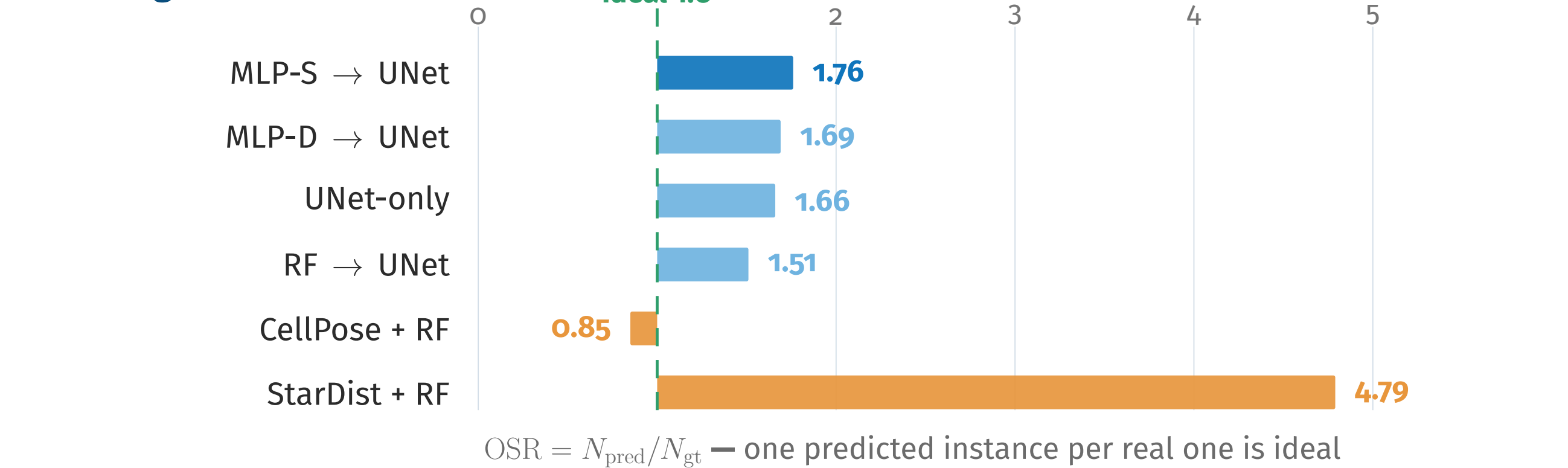
How many organisms are recovered



Where the quality comes from



Over-segmentation



Bootstrapping is what pays: the shallow-MLP pipeline beats direct U-Net training on the same three crops by **+0.037 PQ** and **+0.035 recall**. **Limitation:** an OSR of 1.76 shows the higher recall comes with substantial over-segmentation.